Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology



Course Code
ECE 3112

Course Title
Microprocessor, Assembly Language & Interfacing Sessional

Experiment Date: April 22, 2024,
Submission Date: April 29, 2024

Lab Report 1, 2 & 3: Using Assembly Language to Program 8086 Microprocessor to do Different Tasks.

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Conversion From Lowercase to Uppercase.

Tools Used

- emu8086 - 8086 Microprocessor Emulator
- MacTeX -LaTeXcompiler.
- VS Code with LaTeXworkshop extension as a text editor.

Process

Code:

```
DATA    SEGMENT
    CR EQU 0DH
    LF EQU 0AH
    MSG1 DB 'ENTER A LOWERCASE LETTER: $'
    MSG2 DB CR,LF, 'IN UPPER CASE IT IS: '
    CHAR DB ?, '$'
DATA    ENDS
CODE    SEGMENT
    ASSUME CS:CODE, DS:DATA

START:
    MOV AX, DATA
    MOV DS, AX

    LEA DX, MSG1
    MOV AH, 9
    INT 21h
    MOV AH, 1
```

```
19      INT 21h
20
21      SUB AL, 20H
22      MOV CHAR, AL
23
24      LEA DX,MSG2
25      MOV AH, 9
26      INT 21h
27
28      MOV AH, 4CH
29      INT 21h
30
31  CODE    ENDS
32
33  END START
```

Output

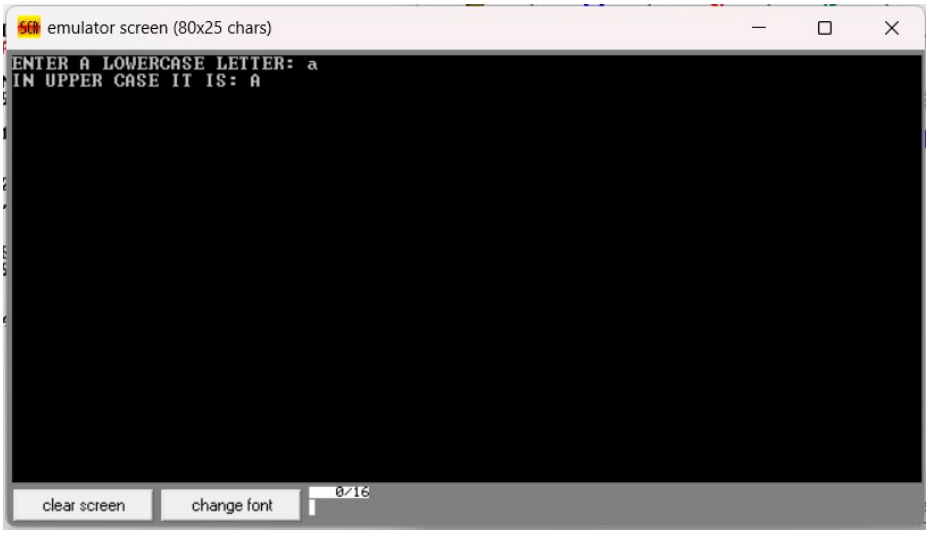


Figure 1: Lowercase to Uppercase

Conversion From Uppercase to Lowercase.

Tools Used

- emu8086 - 8086 Microprocessor Emulator
- MacTeX -LaTeXcompiler.
- VS Code with LaTeXworkshop extension as a text editor.

Process

Code:

```
    DATA    SEGMENT
    CR EQU 0DH
    LF EQU 0AH
    MSG1 DB 'ENTER A UPPERCASE LETTER: $'
    MSG2 DB CR,LF, 'IN LOWERCASE IT IS: '
    CHAR DB ?, '$'
DATA    ENDS
CODE    SEGMENT
    ASSUME CS:CODE, DS:DATA

START:
    MOV AX, DATA
    MOV DS, AX

    LEA DX, MSG1
    MOV AH, 9
    INT 21h
    MOV AH, 1
```

```
19      INT 21h
20
21      ADD AL, 20H
22      MOV CHAR, AL
23
24      LEA DX,MSG2
25      MOV AH, 9
26      INT 21h
27
28      MOV AH, 4CH
29      INT 21h
30
31  CODE    ENDS
32
33  END START
```

Output

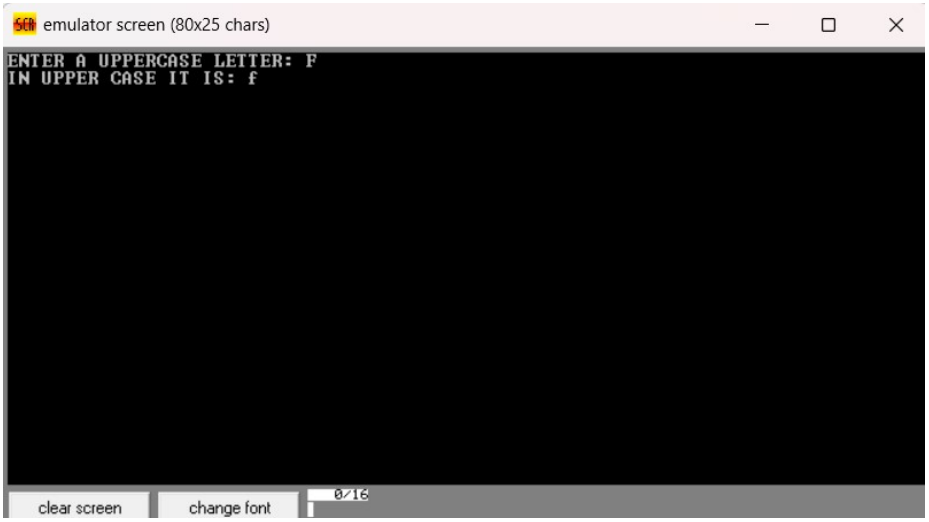


Figure 2: Uppercase to Lowercase

Adding Two Numbers

Tools Used

- emu8086 - 8086 Microprocessor Emulator
- MacTeX -LaTeXcompiler.
- VS Code with LaTeXworkshop extension as a text editor.

Process

Code:

```
DATA    SEGMENT
    CR EQU 0DH
    LF EQU 0AH

    MSG1 DB 'ENTER FIRST NUMBER: $'
    NUM1 DB ?
    MSG2 DB CR,LF,'ENTER SECOND NUMBER: $'
    NUM2 DB ?
    MSG3 DB CR, LF, 'SUM = '
    NUM3 DW ?,'$'
DATA    ENDS

CODE    SEGMENT
    ASSUME CS:CODE, DS:DATA

START:
    MOV AX, DATA
    MOV DS, AX
```

```
19
20      LEA DX, MSG1
21      MOV AH, 9
22      INT 21h
23
24      MOV AH, 1
25      INT 21h
26      SUB AL, '0'
27      MOV NUM1, AL
28
29      LEA DX, MSG2
30      MOV AH, 9
31      INT 21h
32
33      MOV AH, 1
34      INT 21h
35      SUB AL, '0'
36      MOV NUM2, AL
37
38      MOV BH,0
39      MOV BL, NUM1
40      MOV AH, 0
41      MOV AL, NUM2
42      ADD AX,BX
43      MOV NUM3, AX
44
45      ADD NUM3, '0'
46
47      LEA DX, MSG3
48      MOV AH, 9
49      INT 21h
50
51      MOV AH, 4CH
52      INT 21h
53
54  CODE    ENDS
55
56  END START
```

Output



Figure 3: Uppercase to Lowercase

Keywords

CR, LF Carriage Return & Line Feed. Both together is used to go to the next line. CR is used to go to next line and LF to go to the start of the line.

DB Data Byte. Used to declare type of variable.

MOV Move. Used to move data from the second register to the first.

LEA Load Effective Address. Used to finding address of variable, and addition or subtraction without affecting flags.

INT 21h Interrupt. Used to give emu8086 take back control.

SUB Subtract. Used to subtract the second value from the first one and store in the first one.